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RESEARCH-ARTICLE

An Introduction to Weave Structure for HCI: A How-to and Reflection on Modes of Exchange

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An Introduction to Weave Structure for HCI: A How-to and Reflection on Modes of Exchange

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Abstract

As HCI continues to integrate craft techniques into its repertoire, tensions emerge between what is new and known, knowledge that resides in communities and histories rather than individuals, and how to transfer between the written word and material know-how. We explore these tensions through the process of writing, instructing, and create an introduction to weaving force sensors. The goal of this project is two fold: first, to help HCI understand the potential that a deep understanding of weave structures can hold for advancing our field, and second, to explore how a pictorial might support formats that have long been used for communicating craft knowledge.

Authors Keywords

Weaving; Woven Structure; Activity Book; E-textiles.

CSS Concepts

- Human-centered computing

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“From aeronautical altitudes, the crisscross grids of earth’s cities seem to be two-dimensional planar arrangements, as do woven fabrics, seen from a distance. Seen from inside cities or within the loom, both cities and fabrics disclose [omnidirectional], multidimensional structure of great complexity.”
- Lois Martin quoting Buckmeister Fuller’s letter to Anni Albers[19]

Introduction

Textiles describe more than cloth, they are systems of socio-material-cultural calculus that have captivated minds for centuries. They are complex mechanical systems composed of fibers—spun, interlaced, looped, and otherwise tangled into each other in more or less controlled ways. This pictorial is a love letter to weaving written to infect the minds of HCI practitioners with a desire to walk through cloth, travel and feel the paths of the yarns, and illuminate a seemingly endless space of design possibility (interactive and otherwise). Textiles can support many applications within HCI and design, yet, there are also several challenges to framing woven knowledge as a traditional “contribution” to the field [11,21]. For example, while researchers may well discover new forms of woven sensors, communicating specific relationships allowing for that kind of sensing in ways that are understandable, replicable and adaptable to other applications can be very difficult. It’s one

challenge to design a novel textile sensor, it’s another to publish a paper where you must also teach the reader the fundamentals of the craft that created that sensor. So, focusing only on what worked leaves out the knowledge that allowed “working” to emerge as a possibility.

Joining with a body of researchers interested in pushing the form of the pictorial towards more instructive and active modes of engagement (e.g. [26,33]) and others interested in demystifying weaving [24], we use the form of a pictorial to present our research through the mode of historical craft publications. How can craft texts help us communicate the know-how, curiosity and joy that underpins this growing area of HCI inquiry? We look to craft publications such as the exchange of newsletters, instruction books, encyclopedias, and guild journals and apply their techniques to the area of developing woven force sensors, attempting to articulate the multitude of factors in their construction that can be tuned for an application. Thus, we see the form as a way to communicate a series of parameters and guidelines for success alongside specific recipes of techniques that worked. We see benefits in this project for researchers intending to use weaving to produce interactive materials (e.g. [9,30]), practitioners in the area of creative coding and interactive fabrication [1,2,5,6,8,36], and computer aided design (e.g. [12]) as the underlying calculus of woven structure and pattern offer a rich set of computational interface inspirations (e.g. [20,32]).

What follows below is a publication within a publication. Pages 3-11 can be folded into a handbook to be used as both reference and activity book. The workbook offers itself as a compass, to introduce the fundamental wayfinding, “textile thinking” [10] or perhaps “drifting” [13] techniques to HCI practitioners along their journey to discover new forms of textile (specifically woven textile) based interactive applications.

SITUATING CRAFT KNOWLEDGE

We look to craft textbooks, guild publications, encyclopedias and handbooks for insights into how craft knowledge is represented and distributed. We draw from these texts to communicate the foundations of a weaving practice in a way that can best communicate practical and replicable knowledge to a broad community. Of course, we complement this knowledge with techniques that were passed by word of mouth and from person to person or learned by doing that can’t be easily cited. Weaving texts demonstrate that each weaver’s vocabulary, conventions, and practice is finetuned by their local community or the traditional and personal practice through which this knowledge developed. Our resources all draw from western and Eurocentric traditions of weaving and we must acknowledge that we are leaving out forms of weaving prominent in the global Souths and indigenous traditions that hold their own incredible value. The personal and community-oriented nature of weaving often leads to a lack of language standards. Thus, these texts feel modest in that they simply communicate what a particular weaver did, rather than a set of recommendations for what should be done. Furthermore, these texts include activities and lessons intended for the reader to not only read but embody through action. By drawing from these texts, we intend to move this ethos into the realm of academic publishing—carving a space to consider the pictorial as a platform for sharing activities and modes of representation that can scaffold craft knowledge



Inspirational texts for this project included: A Complete Book of Drafting for Handweavers [16], On Weaving [3], The Manual of Swedish Handweaving [7], Mastering Weave Structure [4], and the Complex Weavers Journal (Guild-Members Publication) [34]. The forms of inspiration we drew from each are listed with each image.

SITUATING WOVENS in the TEXTILE LANDSCAPE

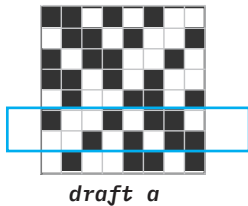
Textiles are made from fibers, materials whose length to width ratio is very high. Fibers come from any number of natural or synthetic materials including animal hair, plant matter, extruded plastics, metal, etc. These fibers are processed in multiple ways before we encounter them as yarn, and then as yarn, become embedded in even more complex structures. Yet, if one were to desire, a smart textile could be tuned, so to speak [4], at every step beyond fiber. The resulting cloth tends to be classified into three categories: woven, knit, or non-woven depending on the methods of production and resulting physical structure, though hybrid categories also exist (e.g. see the knit-weave [35] structure or, later in our work, the embedding of a non-woven felt structure into a woven pocket structure). Our primary contribution to this research is in the area of weaving, as we take a very deep exploration into woven structures. Our hope is that a researcher could read/do the activities offered in this paper before embarking upon or reviewing

a paper about woven electronics and be sufficiently up to speed to understand, in both cognitive and embodied ways, how the technique functions.

As a material for interaction and form-giving (in 2 or 3 dimensions), textiles have been shown to offer novel modes of design and construction: Textiles are inherently multi-material, one can combine fibers of different types prior to spinning yarn, or can mix fibers and filaments in knitting, weaving, and spinning (e.g. [23]); Textile structures are adhesiveless and can be assembled, disassembled, and mended (e.g. [18,31]); textiles are highly customizable (e.g. [15,27]) and processes of weaving, felting, and knitting can be used to directly give objects 3D forms without the use of sewing (e.g. [17,28,29]). Considering the breadth of applications within and outside of HCI, it’s possible interest in using textiles structures for interaction will grow. This work can serve as a primer for such growth in the area of weaving.

TRY THIS.

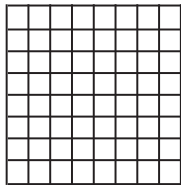
Test your knowledge after reading by answering these questions.



1. Draw the cross section of the draft rows outlined in blue

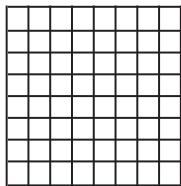


2. Is this structure single or double layered?



3. Draw the inverse of *draft a* in the grid to the left

4. Which of the three core structures was *draft a* derived from?



5. Create a two layer weave draft that has structure *a* on top, and structure *b* on the bottom layer



6. Draw a draft corresponding to the cross-section below



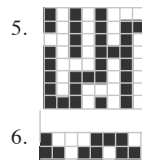
answer key



2. single, the interlacements in the center prevent the structure from being pulled into two separate layers



4. this draft has several twill sections that have been rotated around a central point.



fold --- front and back covers

WEAVING AND DRAFTING FORCE SENSORS

This page represents the front and back covers of the handbook. Fold pages 4-11 along the dotted lines and insert them into the "spine" created by this page. Sew or staple for structure and keep as a handy reference when you encounter weaving in research or decide to start playing with weaving yourself. The activities on the back cover consist of a quiz to test your textile knowledge.

This book intends to help new or beginner weavers learn more about how woven cloth can be of use in interaction design.

We focus specifically on the variable structures that can be achieved in woven cloth and the implications of those structures for interactive applications. Specifically, we draw from three years of research learning weaving and refining structures that act as resistive force sensors. We attempt to draw this line through the handbook by focusing on force sensing specifically, but feel strongly the general themes behind these specific insights can form the grounds for experimentation in other areas of sensing and actuation.



separate cover
(this page) from
internal pages
(pages 4-11)



fold cover

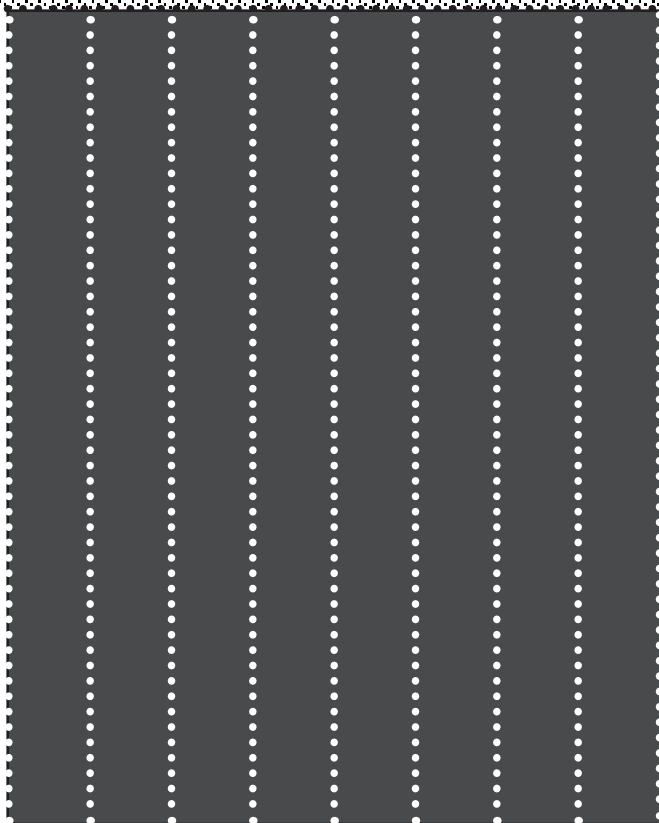


fold each internal
page and stack
folded pages atop
each other



assemble, place
internal stack
inside cover. bind
as desired

<--- insert wefts this way

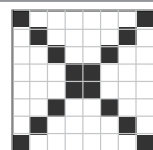


this is your "loom"

fold -- spread 1

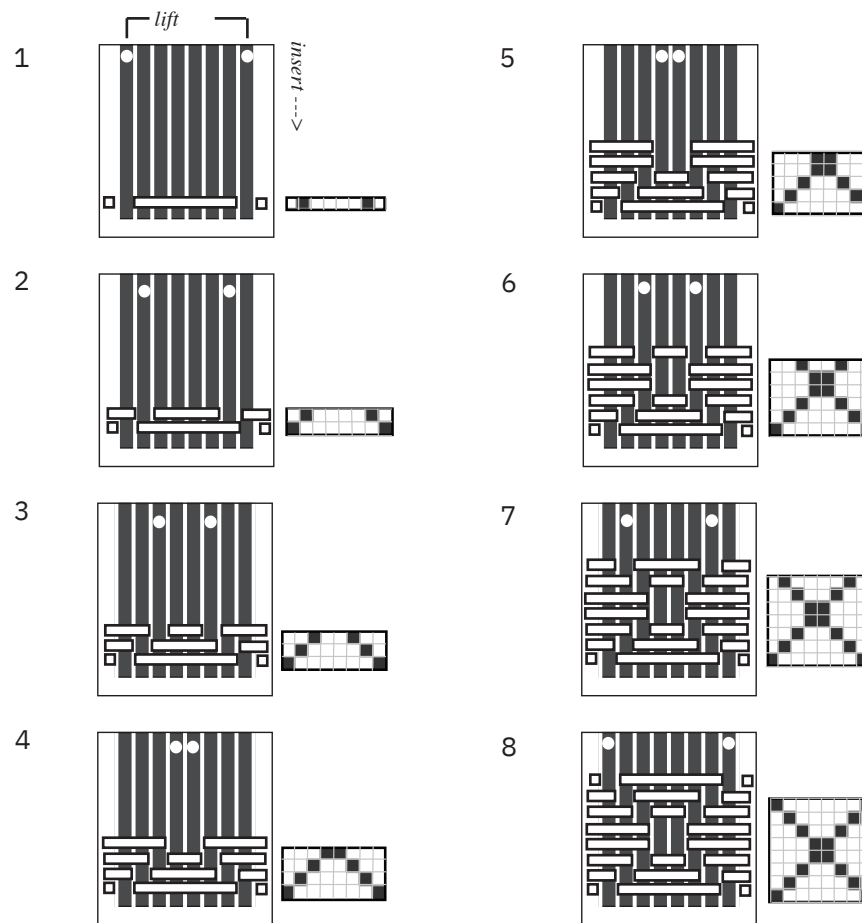
LET'S FOLLOW A WEAVE DRAFT

The pattern a weaver follows to create cloth is called a **DRAFT**. It is a binary map or plan that tells them which warps/heddles to lift on each pick. A  cell represents a raised warp, a  cell represents a lowered warp. Our draft is followed from the bottom up.



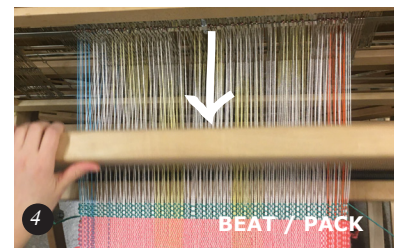
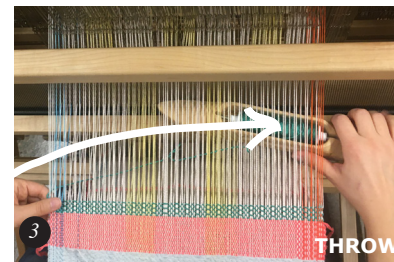
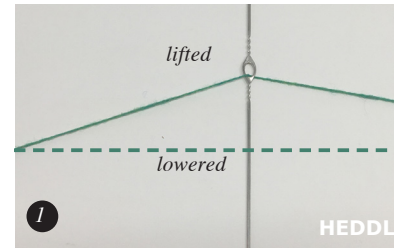
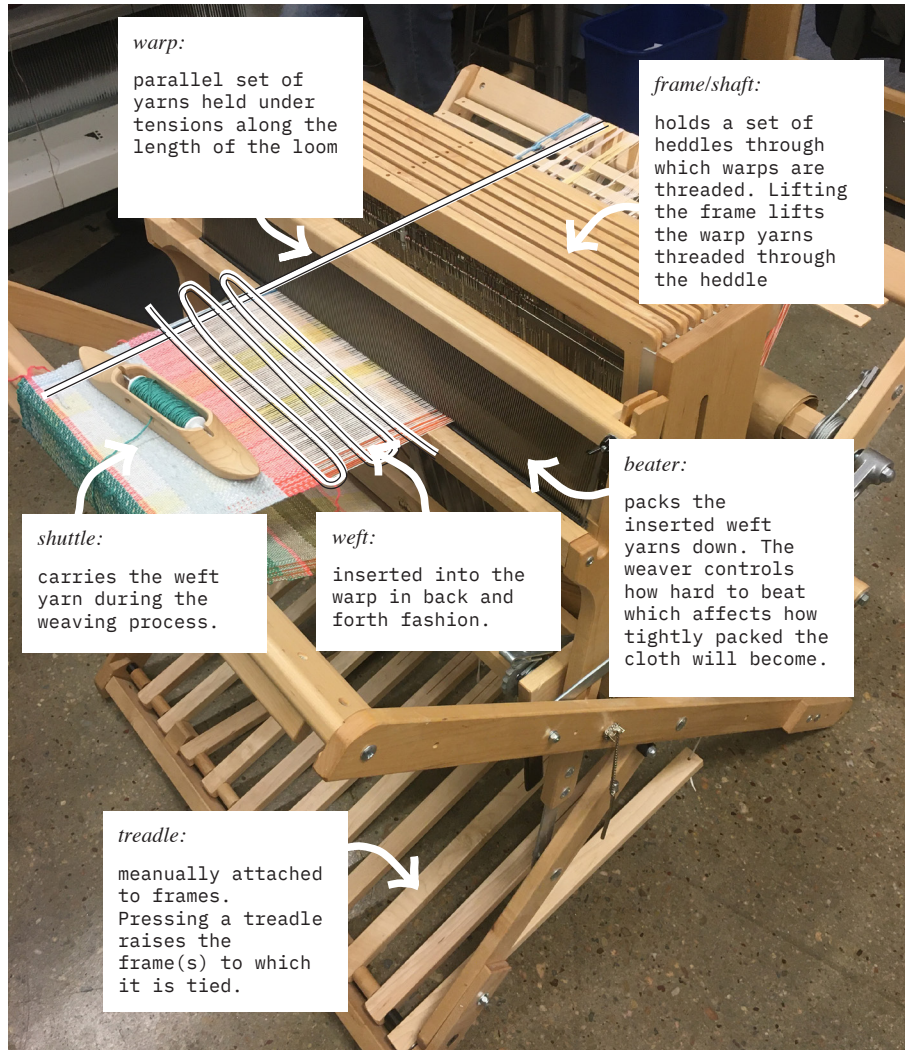
TRY THIS.

Cut out the paper loom on the left and we'll weave a draft together. Lift the black warps and insert the wefts as indicated below



THE HANDWEAVING PROCESS and TERMS TO KNOW

Understanding the structural possibilities of woven cloth depends on a deep understanding of the processes by which woven cloth is assembled. While, as Albers writes, “Any weaving, even the most elaborate, can be done, given time, with a minimum of equipment” [3], we focus on a class of equipment used primarily in industry and western weaving style: the horizontal or two-beam weaving **LOOM**.



The images on the left and above depict a **HARNES LOOM** where the pattern is “programed” into the loom by threading heddles on different shafts, which are lifted and lowered using treadles while weaving.

The following cycle is repeated during the process of weaving:

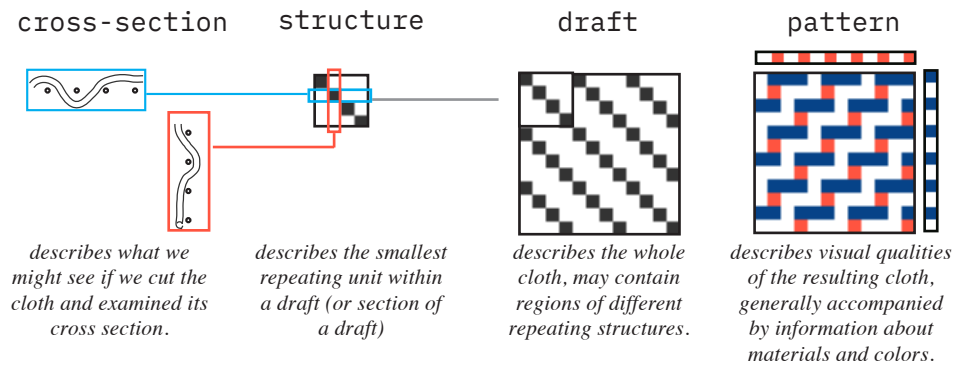
1. Following a **DRAFT**, the weaver raises a selection of warp threads. On your paper loom, you did this by hand. On a computerized jacquard loom, a bitmap image tells the machine which heddles to raise and lower just like you used the black and white cells to determine what to lift or not.
2. When heddles are lifted, they raise the warp threads that are threaded through them and create a **SHED**, which is the term for the space between the raised and lowered warp threads.
3. A **SHUTTLE** carrying the weft thread is **THROWN** through the shed, adding a weft row, or **PICK**, to the cloth.
4. A weaver closes the shed (lowers all warps) and **BEATS**, pressing the new weft on top of the previous weft. This pressing **PACKS** the weft yarns in the weave
5. Repeat 1-4



Many of the samples created in this book were made using the computer controlled **JACQUARD LOOM** show above. Jacquard looms allow for each heddle to be lifted independently of the others. This loom is a TC 2, contains 30 warps per inch and is 43” wide. It is programmed with bitmap images.

DESIGNING WOVEN STRUCTURES

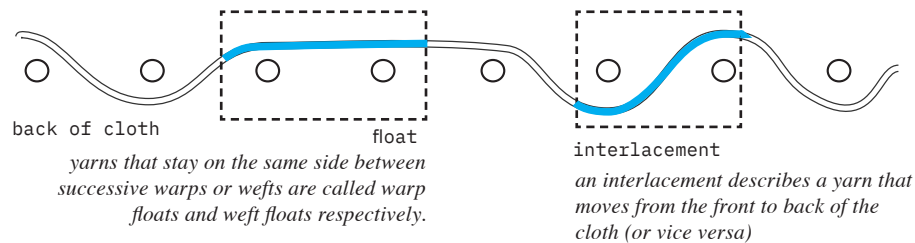
Woven cloth can be understood in multiple ways—a mechanical system, a multi-material assembly, a textured image—to name a few. The images below show four different representations of the same woven cloth. Each representation makes different qualities of the cloth more readily visible. Representations are closely interrelated, for instance, changing the draft can change both the pattern and structure. Because we are interested in sensing, and sensing is possible when a mechanical deformation of a material leads to a change in its electrical characteristics, we focus more deeply on structure as it is represented by draft and cross-section.



DEFINING FLOATS AND INTERLACEMENTS

The combination of floats and interlacements determines the qualities of the cloth. For example, longer floats can create denser and softer cloth while more interlacements create more rigid and thin structures. We believe that the cross-section makes it easiest to visualize floats and interlacements and therefore the most helpful when focusing on the mechanics of the cloth.

front/face of cloth circles represent warp ends, curve line represents the movement of the weft yarn in and out of the warps



for reference, the cross section would be representing the following draft row



FLOATS ARCH and STACK, INTERLACEMENTS SPREAD APART

As you weave pick by pick and beat the weft in between, floats and interlacements push and pull on each other to shape the fabric in three dimensions. Interlacements will constrain the movement of the weft and create air and space between the crossing warp and weft. Floats are mobile and move like a bezier curve, arching outward from the surface of the cloth. When considering force sensing, in which you want yarns to touch proportional to force, this arching is useful in keeping the yarns separated when no pressure is applied, and in close contact with pressure.



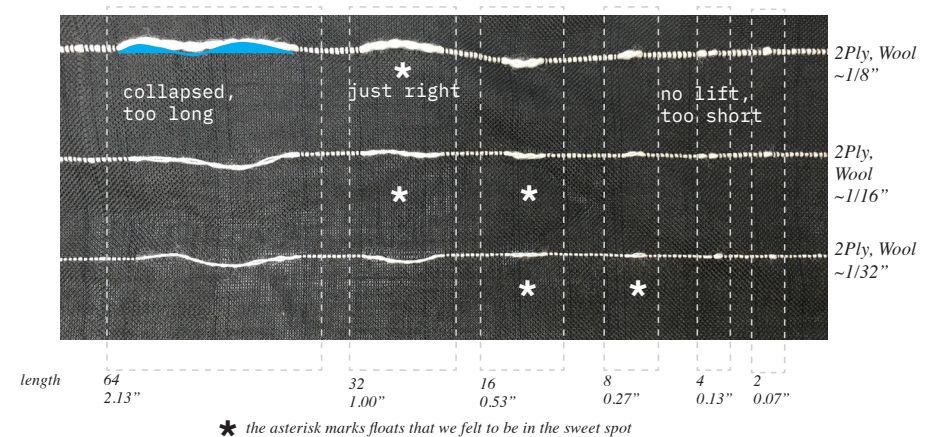
fun with floating steel yarn and magnets



floats of a length specific to each material will spring back to shape after being pressed. We call this the sweet spot.



We aimed to identify a sweet spot, in which the arch is maximized and consistent. Floats shorter than the sweet spot do not contain much vertical lift from the surface and floats longer than the sweet spot tend to collapse and rest upon the surface of the cloth. The sweet spot will vary for each material a weaver chooses as well as the setup of their loom. The best way to identify this region is to test different float lengths on your loom and with your desired materials but in general, thicker yarns will have longer sweet spots than thinner yarns of the same material.



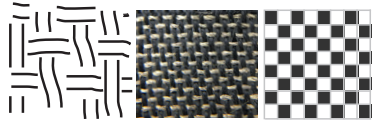
fold --spread 3

DESIGNING STRUCTURES VIA DRAFTING

The first approach to designing woven structures is to design the draft. In this process, weavers typically begin with a core structure (left) and apply different manipulations (right) or derivatives to it to generate new structural variations.

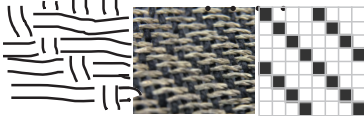
CORE STRUCTURE

tabby



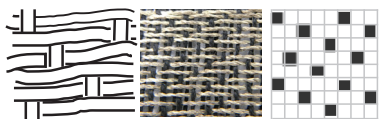
Tabby describes a structure composed entirely of interlacings. In a balanced cloth, this creates the most space between weft yarns, creating a light and thin cloth. The air between wefts means that each weft is electrically insulated.

twill



Twill offsets interlacings in subsequent picks by one. This allows yarns to more tightly stack atop each other, creating more durable and slightly thicker cloth. The wefts are electrically insulated unless the cloth is pulled diagonally in the direction of the twill.

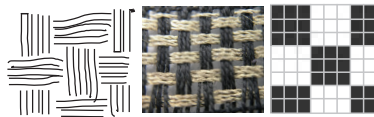
satins



Satin maximizes the distance between interlacings on subsequent picks creating thicker cloth and a tighter stacking than twill. The wefts are electrically insulated unless the fabric is pulled in either diagonal.

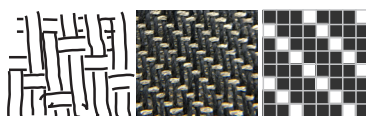
DRAFT MANIPULATION

enlarge



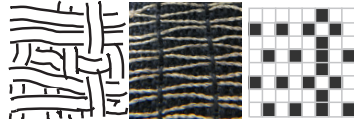
Basket weave is an enlarged version of tabby where warps and wefts interlace every n th row and column. The structure above is a 3/1 basket weave. Note how the enlarged floating regions function like a single thicker yarn.

invert



Inverting a structure swaps all weft floats to warp floats and vice versa. Put another way, inverting moves all the weft floats to the back face of the cloth. This twill is said to be "warp facing" as there are more warp floats than wefts on the face of the cloth.

interlace



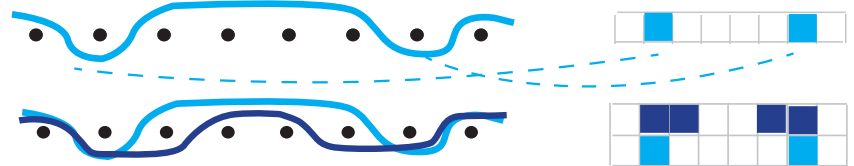
Interlacing takes two structures and interlaces their rows together. For example, if you want your satin weave to be thinner, you might interlace it with tabby. Tabby adds space between wefts, where the satin adds floats above the surface of the cloth.

DESIGNING STRUCTURES VIA CROSS-SECTION

We can also focus on structure more centrally in design by drawing the cross sections of your cloth and then translating those into draft.



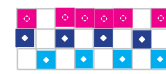
The circles represent each warp. We added perspective lines to help the viewer visualize the direction from which we are



Draw each weft yarn in the pattern you'd like it to interlace with the warp yarns and move between the faces of the cloth. Then, fill in a square for each warp yarn the weft moves under in the draft.

INTERESTING CROSS-SECTIONS TO EXPLORE

place a yarn to one side of the cloth



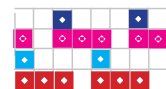
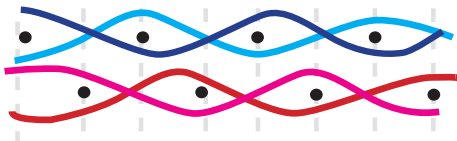
this structure places the pink yarn on the back surface of the cloth

create cut or opening in the cloth



only weaving half the width of the loom with two different yarns can create openings.

weave two layers at once

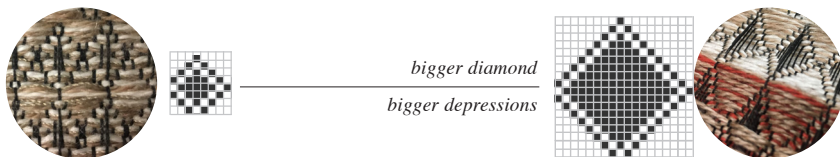
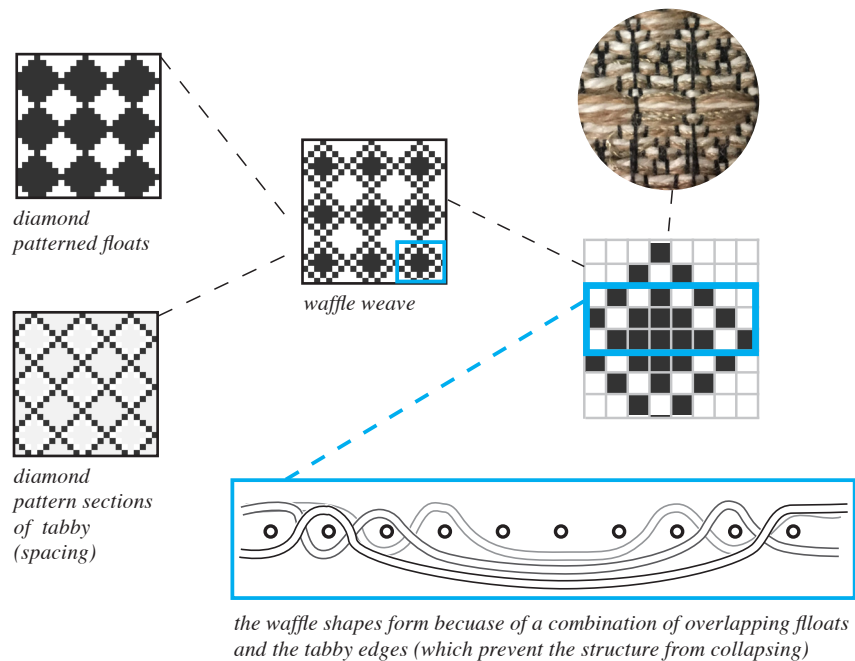


creates two layers atop each other. Here, we move the circles representing warps to two rows to more easily visualize the layers

FORCE SENSITIVE WEAVE STRUCTURES 1: WAFFLE WEAVES

The first class of force sensors emerge from a class that Cyrus-Zetterström describes as “Structures that Form Uneven Surface Textures and Openings” and include the structures classified as cord, honeycomb, waffle weave and crepe weaves to name a few. A waffle weave is a structure with floats organized into large diamond shapes, bound by sections of tabby weave between the diamonds. “The long warp and weft floats tend to draw the fabric together, while the closely woven tabby areas tend to extend it. In this way, ridges and depressions are formed” [7].

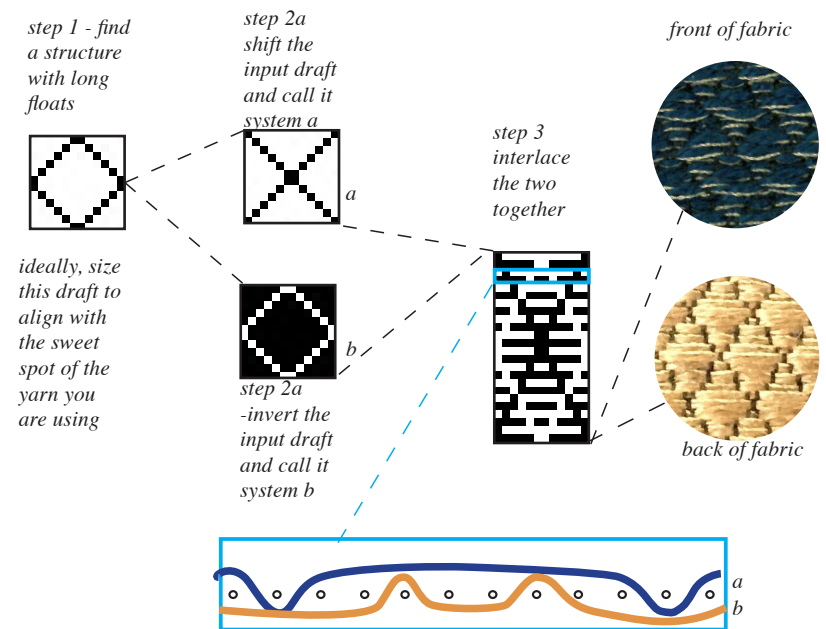
how to make a waffle structure



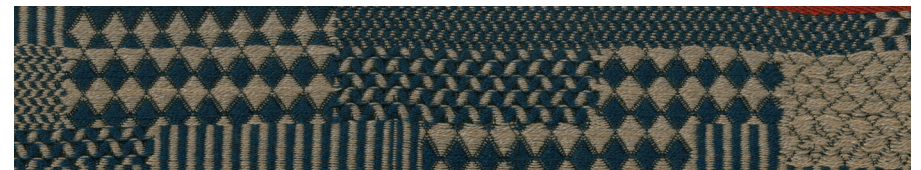
FORCE SENSITIVE WEAVE STRUCTURES 2: MULTI-PIC STRUCTURES

The second class of force sensing structures are multi-pic structures. It's useful to think of these structures as having two weft systems or two structures that are interlaced into each other. For simplicity's sake, consider a weft system as describing wefts of a particular color, let's say blue and tan. Two pic structures ensure that the weft system a, or the blue threads and weft system b, the tan threads, tuck in ways that place a/blue on one side of the fabric, and b/tan on the other. Using this technique, one can create bold 2-color graphic patterns. From a sensing perspective, you can create fabrics where there are several small pockets, almost like bubble wrap.

how to make a 2-pic structure

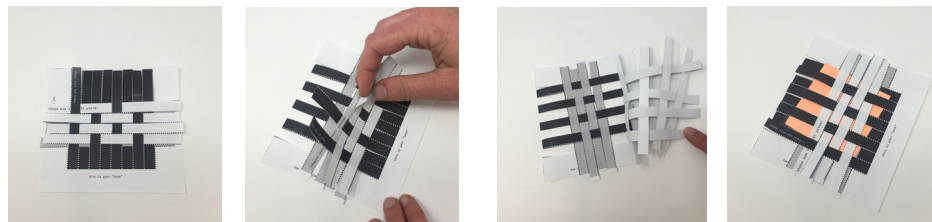
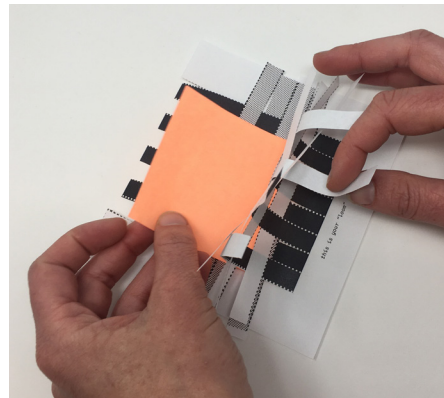
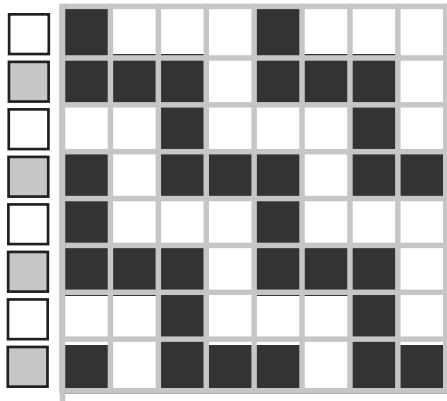


An assortment of two pic structures in blue and tan. Each derived from different starting structures and following the process above.



FORCE SENSITIVE WEAVE STRUCTURES 3: MULTI-LAYER STRUCTURES

Multilayered structures have already captivated the HCI audience in that they are quite simple to formulate and lead to a comfortable mapping between existing cognitive models of sensors as laminated or multi-layer systems and woven structures (e.g. [9,30]). Yet, there are aspects of multi-layered weaving that have yet to be explored, such as the binding of layers, the introduction of pockets, or the ability to “inlay” patterns on each layer. The most important rule when considering multilayer drafts is that every layer will reduce the warp-resolution (calculated in warp ends per inch) by half. Thus, if your loom is warped at 30 ends per inch, a two layer weave will have each layer woven at 15 ends per inch, and a 5 layer weave would have 6 ends per inch.



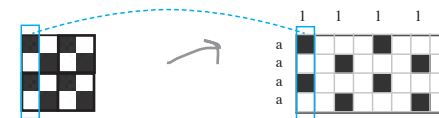
TRY THIS.

Follow this draft on your paper loom. Alternate grey and white on each pick. If you pull up a warp you will be able to pull up the entire first layer. Now you can insert something (perhaps a sensing something) in between the layers.

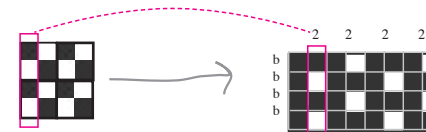
fold --spread 6

A FORMULA FOR MULTI-LAYER WEAVES

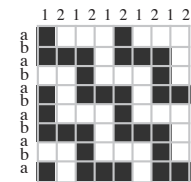
Weaving has its own unique calculus which allows us to follow a formula to draft multi-layer structures. This involves dedicating particular warps to particular layers and then mapping the structure onto those layers.



To give the top layer a tabby structure, map the structure to the warps assigned to layer 1.



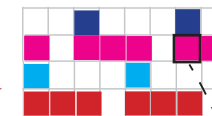
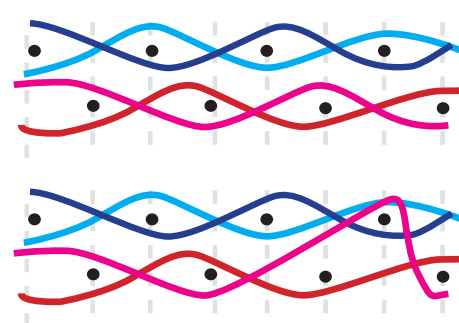
To give the bottom layer a tabby structure, map the structure to the warps assigned to layer 2 and lift the warps assigned to layer 1.



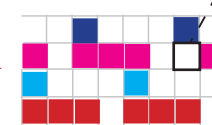
interlacing the two drafts together results in a two-layered cloth with a tabby structure on both layers

JOINING LAYERS

At times, you may want to weave a two layer fabric but have it function as a single layer in some regions and a hollow layered structure on another. This can be accomplished with a technique we called joining.

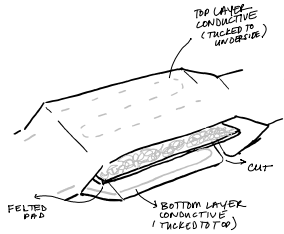


start with the two-layer tabby structure we introduced on spread 4.



modify at least one layer to interlace into the other layer under a weft float. If only sparsely interlaced, the binding yarn will be mostly invisible.

FORCE SENSITIVE WEAVE STRUCTURES 3: MULTI-LAYER STRUCTURES

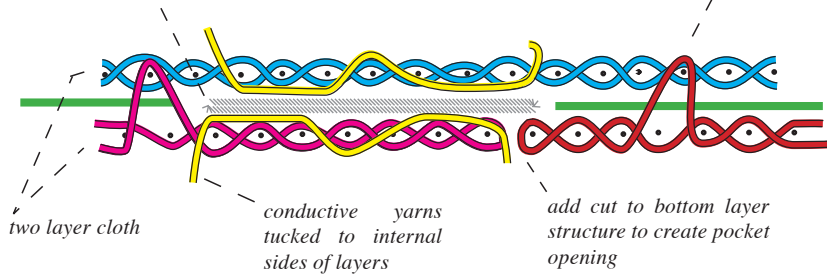


The force sensors that we believe offer the best performance took a structure which integrates techniques introduced on the previous page and integrates a non-woven structure. Two conductive pads are embedded on the inside of the separate layers. We sandwich a resistive felted pad between. The fibers in the felt compress on press and expand back on release. We hand felted our pads so that they would be very airy but consistent. This assembly is not purely woven, instead, it uses the weave

as a kind of custom container or circuitry structure for the felted steel inside the pocket. This offered advantages for speed of production and opportunities for reuse via modular design. Furthermore, it created quite robust force sensing values. This suggests that multiple fiber structures can play well together, and that knowledge of one might be complemented by knowledge of another.

insert a conductive felt pad as well as insulated wiring through center layers

added stitch to join layers in non-pocket regions



NEEDLE FELTING SENSOR PADS



Felt is a non-woven structure that entangles fibers directly into cloth rather than passing through a "yarn" step. This structure complemented our woven sensing by allowing us to quickly make robust sensor pads to slip in wired pockets. We created sensor pads by needle felting a mix of roughly 80% merinox roving and 20% churro wool roving (to add air and structure). We needle felted until the pads were about 1/4" thick at rest and that they would not rip when pulled.



photos of the front and back sides of our multi-layer structure

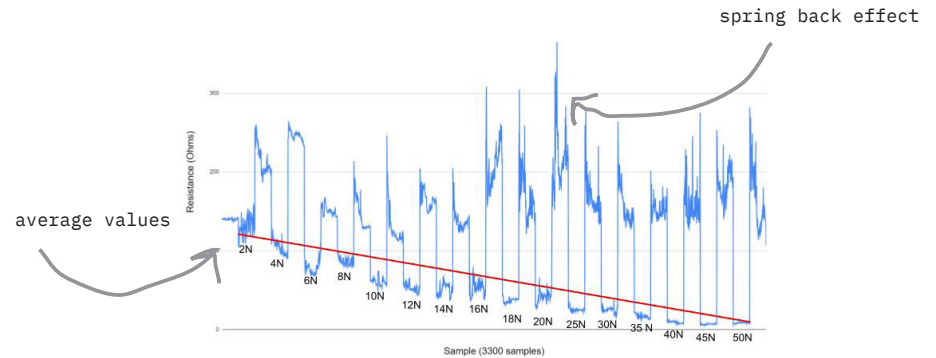


details of the conductive yarns peeking to the surface at the edges of the pocket



detail of the felt pad inserted inside the pocket opening

SENSOR PERFORMANCE



We characterized this force sensor using a digital force gauge and an Arduino code to measure output voltage. These tests revealed a correspondence between the force exerted on the fabric and the associated resistance change. After the press was released, we observed a bounce back effect which we attribute to the inherent spring of the felted wool. Overtime, the fibers may compress, in which case, we can replace the old pads with new pads by simply inserting them into the pocket.

LESSONS LEARNED MAKING A FABRIC THAT REMEMBERS

The techniques outlined in this workbook were all discovered and developed through the development of “A Fabric that Remembers.” This fabric is inspired by the inherent memory of yarns (to hold shape, smell, etc) and forms of memory imparted through computation (e.g. data stored). It aims to bring attention to the multiple ways fabrics hold memories. A Fabric that Remembers is a woven cloth with six force sensitive regions which, when pressed, transmit press data to a cloud server. A web-based user interface reads from the server and updates the pressure collected on a device. It has modes to show press data in real time as well as records of presses over time.

The cloth, in its current form, has developed over the past three years. We took inspiration from a structure demonstrated by Kobakant [37] that used waffle weave and began to search for structures with similar mechanical qualities. Part of this search involved chance by making a habit of throwing a conductive yarn with each pick of any draft we wove to see if the resulting cloth would have interesting sensing properties. When doing this on a file of swatches left by a former artist in residence, we discovered that multi-pic structures offered sensing as well as opportunities to use bold colors and visual patterns. This opened our eyes to the realization that the design space was much wider than we originally anticipated (see lesson 1).

We turned our attention to developing 2-pick structures that could support force

sensing. Because thinner conductive threads are more plentiful and conductive than thicker conductive yarns, we created our sensors by throwing both a conductive thread and thicker non-conductive yarn on each pick. We found that the ratio between these yarns determines just how much the conductive yarns will touch (or not) to create the changes in resistance required. We found a good match between a 2/10 weight woolen weaving yarn and conductive thread (234/35 statex silver coated nylon) as the wool supports the structural and color qualities of the cloth where the resistive thread allows for the sensing capabilities. After discovering the structures and yarn-width ratios that worked, we looked for a particular yarn that would bring aesthetic value to the project (see lesson 2). We found a tencel/merino blend in the approximate size we needed that offered an airy structure, lustrous sheen (that would emphasize the texture of the fabric), and a wide variety of color-ways we could explore. We used these to make A Fabric that Remembers (version one), which was included in a major interactive exhibition which hosted roughly 42000 visitors. When we got the fabric back, it no longer worked and we determined that the silver coating on the yarns wore off (lesson 3). Furthermore, the connections between conductive yarns in the fabric by knotting had become loose or entirely pulled apart.

To remedy our fabric, we needed a new structure that did not rely on the resistive yarns we originally used. In parallel

Lesson 1: Create opportunities for chance discoveries

*samples created by chance and their
resulting resistance measurements*



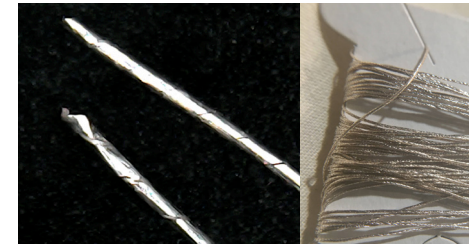
Lesson 2: Fine tune the ratios between yarn-widths prior to making final material selections.

*turning the ratio of wollen yarn to
resistive yarn. Approximate scale between
yarns shown on right*



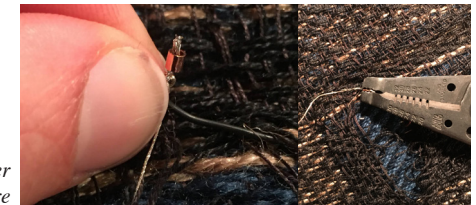
Lesson 3: Use pure metal threads or foil wrapped threads (or test the durability of the conductive coatings).

*Looking at Karl Grimm silver yarn through a
microscope reveals it to be foil wrapped*



Lesson 4: Use crimp beads to reinforce connections between conductive yarns

*Applying a crimp bead between silver
thread and silicone coated wire*



experiments, we saw good sensing results from conductive felt made from stainless steel and merino wool fiber blends. This led us to envision a pocket structure detailed on the previous page. We added pocket openings on the backside of the fabric to allow the felted pads to be inserted and removed, and supported the pockets by binding the

double weave in areas that were not to be used as sensing pockets. We also began to reinforce all connections with crimp beads, which seem to be more durable and lead to more stable sensor readings. The felted inserts to the pocket created a subtle “puff” in the regions that were to be pressed, leveraging a human’s desire to press something that appears squishy.

DISCUSSION & CONCLUSION

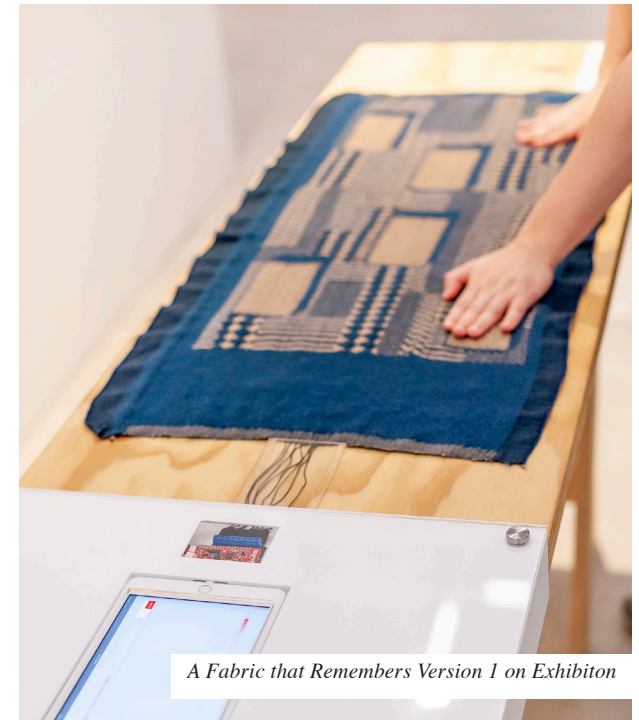
At several points in the development of the pictorial, we were faced with questions about what to tell the reader, how to tell it, and how we might best balance the desire to teach weaving in an embodied sense and demonstrate the application space for weaving in a way that was approachable. We turned to our craft textbooks as a result, often directly mimicking the language, strategies, and narrative structures employed within those texts.

This was a different process than most of our writing for HCI in that it de-emphasized a conversation of novelty and suggestions for applications with a conversation about understanding the non-novel aspects and showing the boundaries within which new applications can be found. By analogy, where novelty-focused papers function like a map, charting the new territory they discovered, this paper functioned like a compass to be used as part of a wayfinding journey through woven structure and sensing.

The evaluation of this format, and its ability to instruct in an embodied way, cannot be fully known at this moment. Ultimately, the reader will decide and, likely, will need to be presented with an occasion. Our immediate goal, then, is to start a conversation for the future about the potential formats that might support craft-based research. While we may not ultimately know if this format is successful, we do feel that the writing process required us to: attend to our material knowledges by accounting for our successes and failures; remake, document and generate diagrams for key steps in the process; and test the activities with fellow lab members who are not-yet weavers. As it is colloquially known, the best way to learn is to teach, and this format allowed us to both teach and relearn our material practice through the lens of instruction and transmission of craft. While we do believe the project presented offers novel insights for both weavers and interaction designers, we wish to shift the focus on the labor, practice, and community that we believe was the locus of novel ideas and techniques.

At the same time, we found ourselves, even in this format, constrained by how much detail and complexity we could provide. We only scratched the surface and would love to make companion volumes that go into more detail about multi-layered structures beyond two layers as well as the variety of ways that fiber and yarn can be designed with unique qualities that become emphasized by the weaving. What became most enjoyable in the process is just celebrating the inherent technicality and ingenuity that has existed in textile production for centuries. For a particular kind of person (ourselves included) the puzzles, calculus and materiality of weaving offers another way of doing science and technology that is deeply embodied, slow, thoughtful, and deeply rooted in community. The sheer vastness of possibility of weaving scaffolds the formation of a collective of travelers through the domain. A discovery is not disruptive, it is constructive to other discoveries, each marking a point of inspiration and ultimately, appreciation for the craftspeople's journey.

Each craft, from embroidery, knitting, pottery, etc, is a vast discipline in and of itself. Researchers in a craft domain face an additional job of explaining the fundamental discipline within which they practice. This leads to shortened descriptions of those disciplines as well as attempts to divert from the past to describe the “new” elements integrated. We believe craft to be an immensely useful collaborator with HCI, yet, to fully integrate it into our practices, we need to envision other modes of communicating these disciplines without our own. Through our pictorial we aimed to demonstrate the possibility of a complementary track for sharing know-how as texts and open up the complexity and wonder inherent in these rich spaces.



A Fabric that Remembers Version 1 on Exhibit

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REFERENCES

- [1] Lea Albaugh, Scott E. Hudson, Lining Yao, and Laura Devendorf. 2020. Investigating Underdetermination Through Interactive Computational Handweaving. In *Proceedings of the 2020 ACM Designing Interactive Systems Conference*. Association for Computing Machinery, New York, NY, USA, 1033–1046.
- [2] Lea Albaugh, James McCann, Lining Yao, and Scott E. Hudson. 2021. Enabling Personal Computational Handweaving with a Low-Cost Jacquard Loom. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, Association for Computing Machinery, 1–10.
- [3] Anni Albers. 1974. *Anni Albers : On Weaving*. Wesleyan, Middletown (Conn.).
- [4] Sharon Alderman. 2004. *Mastering Weave Structures: Transforming Ideas into Great Cloth*. Interweave Press, Loveland, CO.
- [5] alexmclean. Live coding the warp-weighted loom. PENELOPE. Retrieved February 17, 2022 from <https://penelope.hypotheses.org/1780>.
- [6] Kristina Andersen, Ron Wakkary, Laura Devendorf, and Alex McLean. 2019. Digital crafts-machine-ship: creative collaborations with machines. *Interactions* 27, 1: 30–35.
- [7] Ulla Cyrus-Zetterstrom. 1977. *Manual of Swedish Handweaving*. Charles t Branford, Newton Centre, Mass.
- [8] Himani Deshpande, Haruki Takahashi, and Jeeun Kim. 2021. EscapeLoom: Fabricating New Affordances for Hand Weaving. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, Association for Computing Machinery, 1–13.
- [9] Laura Devendorf and Chad Di Lauro. 2019. Adapting Double Weaving and Yarn Plying Techniques for Smart Textiles Applications. *Proceedings of the Thirteenth International Conference on Tangible, Embedded, and Embodied Interaction*, ACM, 77–85.
- [10] Ebru Kurbak Ed. 2018. *Stitching Worlds*. Berlin.
- [11] Raune Frankjær and Peter Dalsgaard. 2018. Understanding Craft-Based Inquiry in HCI. *Proceedings of the 2018 Designing Interactive Systems Conference*, Association for Computing Machinery, 473–484.
- [12] Mikhaila Friske, Shanel Wu, and Laura Devendorf. 2019. AdaCAD: Crafting Software For Smart Textiles Design. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, ACM, 345:1–345:13.
- [13] Bruna Goveia da Rocha and Kristina Andersen. 2020. Becoming Travelers: Enabling the Material Drift. *Companion Publication of the 2020 ACM Designing Interactive Systems Conference*, Association for Computing Machinery, 215–219.
- [14] Julie Holyoke. 2021. *Digital Jacquard Design*. Bloomsbury Visual Arts.
- [15] Cedric Honnet, Hannah Perner-Wilson, Marc Teysier, et al. 2020. PolySense: Augmenting Textiles with Electrical Functionality using In-Situ Polymerization. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–13.
- [16] Madelyn Van Der Hoogt. 1993. *The Complete Book of Drafting for Handweavers*. Shuttle Craft Books, Coupeville, WA.
- [17] Scott E. Hudson. 2014. Printing Teddy Bears: A Technique for 3D Printing of Soft Interactive Objects. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, ACM, 459–468.
- [18] Lee Jones. 2021. The E-darning Sampler: Exploring E-textile Repair with Darning Looms. In *Proceedings of the Fifteenth International Conference on Tangible, Embedded, and Embodied Interaction*. Association for Computing Machinery, New York, NY, USA, 1–5.
- [19] Lois Martin. 2002. The Direction of Cloth: the Horizontal Dimension. *Surface Design Journal* 26, 2.
- [20] David McCallum. 2018. Glitching the Fabric: Strategies of new media art applied to the codes of knitting and weaving. .
- [21] Michael Nitsche and Anna Weisling. 2019. When is It Not Craft?: Materiality and Mediation when Craft and Computing Meet. *Proceedings of the Thirteenth International Conference on Tangible, Embedded, and Embodied Interaction*, ACM, 683–689.
- [22] G. H. Oelsner. 2016. *A Handbook of Weaves*. Forgotten Books, Place of publication not identified.
- [23] Alex Olwal, Jon Moeller, Greg Priest-Dorman, Thad Starner, and Ben Carroll. 2018. I/O Braid: Scalable Touch-Sensitive Lighted Cords Using Spiraling, Repeating Sensing Textiles and Fiber Optics. *UIST 2018*, 485–497.
- [24] Daniëlle Ooms, Nick Voskuil, Kristina Andersen, and Hanna Ottilia Wallner. 2020. Ruta, a Loom for Making Sense of Industrial Weaving. *Companion Publication of the 2020 ACM Designing Interactive Systems Conference*, Association for Computing Machinery, 337–340.

REFERENCES CONTINUED

- [25] Peggy Osterkamp. 2005. Weaving & Drafting Your Own Cloth. UNICORN BOOKS & CRAFTS, Sausalito, Calif.
- [26] Jeroen Peeters, Rosa van der Veen, and Ambra Trotto. 2020. Pictorial Unleashed: Into the Folds of Interactive Qualities. In Proceedings of the 2020 ACM Designing Interactive Systems Conference. Association for Computing Machinery, New York, NY, USA, 925–938.
- [27] Emmi Pouta and Jussi Mikkonen. 2019. Hand Puppet as Means for eTextile Synthesis. Proceedings of the Thirteenth International Conference on Tangible, Embedded, and Embodied Interaction, Association for Computing Machinery, 415–421.
- [28] Timo Rissanen and Holly McQuillan. 2016. Zero Waste Fashion Design. Fairchild Books, London ; New York.
- [29] Michael L. Rivera and Scott E. Hudson. 2019. Desktop Electrospinning: A Single Extruder 3D Printer for Producing Rigid Plastic and Electrospun Textiles. In Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems. Association for Computing Machinery, New York, NY, USA, 1–12.
- [30] Ruojia Sun, Ryosuke Onose, Margaret Dunne, Andrea Ling, Amanda Denham, and Hsin-Liu (Cindy) Kao. 2020. Weaving a Second Skin: Exploring Opportunities for Crafting On-Skin Interfaces Through Weaving. In Proceedings of the 2020 ACM Designing Interactive Systems Conference. Association for Computing Machinery, New York, NY, USA, 365–377.
- [31] Shanel Wu and Laura Devendorf. 2020. Unfabricate: Designing Smart Textiles for Disassembly. In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. Association for Computing Machinery, New York, NY, USA, 1–14.
- [32] Enrique Zeleny. 2013. Ada Deitz Polynomials for Handwoven Textiles. Wolfram Demonstrations Project. Retrieved February 17, 2022 from <http://demonstrations.wolfram.com/AdaDeitzPolynomialsForHandwovenTextiles/>.
- [33] Clement Zheng, HyunJoo Oh, Laura Devendorf, and Ellen Yi-Luen Do. 2019. Sensing Kirigami. Proceedings of the 2019 on Designing Interactive Systems Conference, Association for Computing Machinery, 921–934.
- [34] Complex Weavers. Retrieved February 17, 2022 from <https://www.complex-weavers.org/member-resources/journal-index/>.
- [35] Inlay or Knit-weaving. Assia Brill. Retrieved February 17, 2022 from <https://assiabrill.com/knitting-2/inlay-or-knit-weaving/>.
- [36] Tatsuki Hayama | artworks - Generative Art with Mathematics. Retrieved February 17, 2022 from <https://tomoe.me/generative-art>.
- [37] KOBAKANT. Retrieved February 17, 2022 from <https://www.kobakant.at/?p=432>.